

Week 2 Worksheet: Superposition and Quantum Gates

QCA Fundamentals

Name: _____

Date: _____

Probability Amplitudes and Normalization

1. A qubit is in the state $|\psi\rangle = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$.

- (a) What is the probability of measuring this qubit in state $|0\rangle$?
- (b) What is the probability of measuring this qubit in state $|1\rangle$?
- (c) Verify that this is a valid quantum state by checking the normalization condition.

2. Consider the state $|\psi\rangle = a|0\rangle + b|1\rangle$ where a and b are real numbers.

- (a) If the probability of measuring $|0\rangle$ is $\frac{1}{4}$, what is the value of $|a|^2$?
- (b) What is the probability of measuring $|1\rangle$?
- (c) If a is positive, what are the possible values of a and b ?

Quantum Interference

3. Consider these two quantum states:

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

- (a) Do these states have the same measurement probabilities? Explain.

- (b) If we could somehow "add" these states (after proper normalization), what would the resulting state be? Show your calculation.
- (c) What happened to the $|1\rangle$ components? What type of interference is this?

Quantum Gates

4. Apply the following gates to the given states and show your work:

- (a) $X|0\rangle = ?$
- (b) $H|1\rangle = ?$
- (c) $Z\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) = ?$

5. The Hadamard gate has the property that $H^2 = I$ (applying it twice gives the identity).

- (a) Starting with $|0\rangle$, apply the sequence $H\beta H$. What is the final state?
- (b) Starting with $|1\rangle$, apply the sequence $H\beta H$. What is the final state?
- (c) What does this tell you about the Hadamard gate?

Circuit Analysis

6. Analyze the following quantum circuit step by step:

Initial state: $|0\rangle \rightarrow$ Apply $H \rightarrow$ Apply $Z \rightarrow$ Apply $H \rightarrow$ Final state: ?

- (a) After applying the first H gate:
- (b) After applying the Z gate:
- (c) After applying the second H gate:

(d) What single gate does this entire circuit implement?

7. Design a circuit that converts $|1\rangle$ to $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Explain your reasoning.

Multiple Qubits

8. Consider a two-qubit system where both qubits start in state $|0\rangle$, so the initial state is $|00\rangle$.

(a) If we apply the Hadamard gate to the first qubit only, what is the resulting state?

(b) If we apply the Hadamard gate to both qubits, what is the resulting state?

(c) In part (b), what is the probability of measuring the outcome $|11\rangle$?

9. True or False? Explain your answers.

(a) The state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ can be written as a product of two single-qubit states.

(b) The state $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$ can be written as a product of two single-qubit states.

Looking Ahead: Two-Qubit Gates

10. Next week we'll learn about the CNOT gate, which flips the second qubit if and only if the first qubit is in state $|1\rangle$. Based on this description:

(a) What should happen when CNOT is applied to $|00\rangle$?

(b) What should happen when CNOT is applied to $|10\rangle$?

(c) What should happen when CNOT is applied to $|01\rangle$?

(d) What should happen when CNOT is applied to $|11\rangle$?

11. If we apply CNOT to the state $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$, what do you think the resulting state will be? (This is a preview of entanglement!)

Synthesis

12. Challenge Problem: Suppose you want to create a quantum circuit that takes any input state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and outputs the state $\alpha|1\rangle + \beta|0\rangle$ (essentially swapping the amplitudes of $|0\rangle$ and $|1\rangle$).

(a) What single gate accomplishes this transformation?

(b) Test your answer: If the input is $\frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$, what should the output be?

(c) Apply your gate to verify this works.